

01.

Two strokes, No. g/lm (n) = 1 $N = \frac{1000}{60}$ revs per second.

Diameter = Bore = 110 mm

$$\therefore \text{effective area of piston} = \pi \left(\frac{D}{2} \right)^2 = \frac{\pi}{4} \left(\frac{110 \times 10^{-3}}{2} \right)^2$$

P_i = mean effective pressure = 1×10^5 Pascal

L = stroke = 140×10^{-3} m

$$\therefore I_p = \text{indicated power} = P_i A L N n$$

$$= 1 \times 10^5 \times \frac{\pi}{4} \left(\frac{110 \times 10^{-3}}{2} \right)^2 \times 140 \times 10^{-3} \times \frac{1000}{60} \times 1$$

W.

02

$$I_p = 14.7 \times 10^3 \text{ W}$$

$N = \frac{1000}{60}$ revs per sec

$n = 4$

four strokes engine.

$4D = L$

D is the bore.

$P_i = 5.5 \text{ bar} = 5.5 \times 10^5 \text{ Pascal}$

$$I_p = P_i \frac{A L N n}{2}$$

$$14.7 \times 10^3 = 5.5 \times 10^5 \times \pi \left(\frac{D}{2} \times 10^{-3} \right)^2 \times \frac{4D}{2} \times \frac{1000}{60} \times 4 \times \frac{1}{2}$$

$$\sqrt[3]{\frac{14.7 \times 4 \times 60 \times 2}{\pi \times 5.5 \times 4 \times 4 \times 10^{-4}}} = D$$

$$D = 63.4 \text{ mm}$$

$$\therefore L = 253.72 \text{ mm}$$

03

torque & Brake load = 175 Nm ~~= bp~~
 $n = 1$ $N = \frac{500}{60}$ rounds Per second.
 four strokes

$$b_p = 2\pi NT = 2\pi \times \frac{500}{60} \times 175$$

$$= 9162.98$$

$$b_p = \underline{\underline{9.163 \text{ kW}}}$$

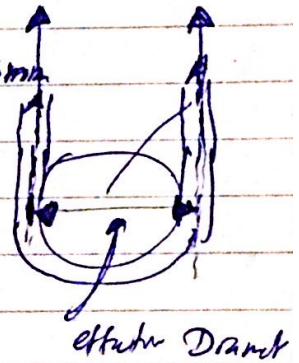
04 $n = 1$, four strokes,

Wheel Diameter = 600 mm rope diameter = 26 mm

$$\text{effective Diameter} = \left(600 + 2 \times \frac{26}{2}\right) \text{ mm}$$

Brake load = 200 N - 30 N = 170 N

$$N = \frac{450}{60} \text{ rounds Per Sec.}$$



$$\text{break power}(b_p) = 2\pi NT \quad \text{--- (1)}$$

$$T = \text{effective diameter} \times (\text{brake load})$$

$$= (600 + 26) \times 10^{-3} \times 170 \text{ Nm}$$

$$\text{Ans (1) } \Rightarrow b_p = 2\pi \times \frac{450}{60} \times (600 + 26) \times 10^{-3} \times 170 \text{ W}$$

$$= \underline{\underline{W}}$$